

Bulan 6: Advanced Features & AI Integration

Minggu 1: GraphQL & Modern APIs

Hari 1: GraphQL Introduction

Project: GraphQLStarter

- Deskripsi: GraphQL API dengan Graphene-Django
- Fitur: Queries, mutations, schema definition
- Keunikan: Interactive GraphQL playground dengan examples

Hari 2: GraphQL Queries & Resolvers

Project: BlogQL

- Deskripsi: Blog API dengan nested queries
- Fitur: Posts, comments, authors dengan relationships
- Keunikan: DataLoader untuk N+1 prevention

Hari 3: GraphQL Mutations

Project: TaskQL

- Deskripsi: Task management dengan GraphQL mutations
- Fitur: CRUD operations, batch updates, optimistic UI
- Keunikan: Transaction support dalam mutations

Hari 4: GraphQL Subscriptions

Project: LiveQL

- Deskripsi: Real-time subscriptions dengan WebSockets
- Fitur: Live updates, filtered subscriptions
- Keunikan: Subscription batching untuk performance

Hari 5: GraphQL Authentication

Project: SecureQL

- Deskripsi: GraphQL dengan JWT dan permissions

- Fitur: Token auth, field-level permissions, rate limiting
- Keunikan: Query complexity analysis

Hari 6: GraphQL Best Practices

Project: GraphQLPro

- Deskripsi: Production GraphQL API dengan monitoring
- Fitur: Query whitelisting, depth limiting, caching
- Keunikan: Automated schema documentation

Hari 7: REST vs GraphQL Hybrid

Project: HybridAPI

- Deskripsi: REST dan GraphQL dalam satu project
- Fitur: Dual API support, unified authentication
- Keunikan: API gateway untuk routing

Minggu 2: Machine Learning Integration

Hari 8: ML Model Integration Basics

Project: MLStarter

- Deskripsi: Scikit-learn model integration dengan Django
- Fitur: Model training, prediction endpoint, versioning
- Keunikan: A/B testing framework untuk models

Hari 9: Image Classification

Project: ImageAI

- Deskripsi: Image recognition dengan TensorFlow/PyTorch
- Fitur: Upload image, classify, confidence scores
- Keunikan: Custom model training interface

Hari 10: Natural Language Processing

Project: TextAnalyzer

- Deskripsi: NLP toolkit untuk sentiment analysis

- Fitur: Sentiment, entity extraction, summarization
- Keunikan: Multi-language support

Hari 11: Recommendation System

Project: RecommendEngine

- Deskripsi: Product recommendation dengan collaborative filtering
- Fitur: User-based, item-based, hybrid recommendations
- Keunikan: Real-time preference learning

Hari 12: Chatbot Integration

Project: ChatBot

- Deskripsi: AI chatbot dengan intent recognition
- Fitur: Natural language queries, context awareness
- Keunikan: Training interface untuk custom intents

Hari 13: Time Series Forecasting

Project: ForecastPro

- Deskripsi: Sales/traffic forecasting dengan ARIMA/LSTM
- Fitur: Historical data analysis, predictions, confidence intervals
- Keunikan: Automated model selection

Hari 14: Model Deployment & Monitoring

Project: MLOps

- Deskripsi: ML model deployment pipeline
- Fitur: Model registry, versioning, monitoring, A/B testing
- Keunikan: Automated retraining triggers

Minggu 3: Advanced Django Patterns

Hari 15: Multi-Tenancy

Project: TenantHub

- Deskripsi: Multi-tenant SaaS architecture
- Fitur: Tenant isolation, shared/separate schemas
- Keunikan: Tenant-aware caching dan routing

Hari 16: Event Sourcing

Project: EventStore

- Deskripsi: Event sourcing pattern implementation
- Fitur: Event log, event replay, projections
- Keunikan: Time-travel debugging untuk data

Hari 17: CQRS Pattern

Project: CQRSApp

- Deskripsi: Command Query Responsibility Segregation
- Fitur: Separate read/write models, eventual consistency
- Keunikan: Automatic read model synchronization

Hari 18: Domain-Driven Design

Project: DDDExample

- Deskripsi: DDD principles dalam Django
- Fitur: Bounded contexts, aggregates, repositories
- Keunikan: Strategic design canvas tool

Hari 19: Feature Flags

Project: FeatureToggle

- Deskripsi: Feature flag system untuk gradual rollout
- Fitur: User/group targeting, A/B testing, rollback
- Keunikan: Feature analytics dashboard

Hari 20: Plugin Architecture

Project: PluginSystem

- Deskripsi: Pluggable architecture untuk extensibility

- Fitur: Plugin discovery, lifecycle management, hooks
- Keunikan: Plugin marketplace dengan ratings

Hari 21: Workflow Engine

Project: FlowEngine

- Deskripsi: Business workflow automation engine
- Fitur: Visual workflow builder, approval chains, SLA tracking
- Keunikan: BPMN 2.0 compatible

Minggu 4: Modern Frontend & Full-Stack

Hari 22: Django + Vue.js

Project: VueDjango

- Deskripsi: Django backend dengan Vue 3 frontend
- Fitur: Composition API, Pinia state, Vue Router
- Keunikan: Server-side rendering dengan Nuxt

Hari 23: Django + Next.js

Project: NextDjango

- Deskripsi: Next.js frontend dengan Django API
- Fitur: SSR, ISR, API routes, optimistic updates
- Keunikan: Edge functions untuk geolocation

Hari 24: Django + Mobile (React Native)

Project: MobileAPI

- Deskripsi: Mobile app API dengan offline support
- Fitur: JWT auth, file sync, push notifications
- Keunikan: Conflict resolution untuk offline edits

Hari 25: WebRTC Integration

Project: VideoCall

- Deskripsi: Video conferencing dengan WebRTC

- Fitur: Peer-to-peer calls, screen sharing, recording
- Keunikan: AI background replacement

Hari 26: Server-Sent Events

Project: LiveUpdates

- Deskripsi: Real-time updates dengan SSE (alternative WebSocket)
- Fitur: One-way server push, automatic reconnection
- Keunikan: EventSource polyfill untuk old browsers

Hari 27: Advanced WebSockets

Project: MultiplayerGame

- Deskripsi: Real-time multiplayer game foundation
- Fitur: Game rooms, state synchronization, lag compensation
- Keunikan: Client-side prediction dan reconciliation

Hari 28: Blockchain Integration

Project: CryptoWallet

- Deskripsi: Cryptocurrency wallet integration
- Fitur: Wallet creation, transactions, balance tracking
- Keunikan: Smart contract interaction (Ethereum)

Hari 29-30: Full-Stack Capstone

Project: LearningPlatform

- Deskripsi: Online learning platform (Udemy-like)
- Fitur: Course management, video streaming, progress tracking, certificates, live classes (WebRTC), Q&A forums, AI recommendations, payment integration, mobile app API
- Keunikan: Adaptive learning paths, peer review system, gamification
- Scope: 2 hari untuk comprehensive e-learning MVP